Determination of the radiation doses and activity of radioactive substances

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The radioactive behavior of a cesium 137 preparation is examined. The local absorbed dose is measured directly at the lead cube, at a distance of one meter, and at the wall of the lab. For the different dose rates, the short-time measurement series yields $\dot{D}_{0\mathrm{m}} = (4.25 \pm 0.12)\,\mu\,\mathrm{Sv\,h^{-1}}$ and $\dot{D}_{1\mathrm{m}} = (0.193 \pm 0.030)\,\mu\,\mathrm{Sv\,h^{-1}}$. From the long-term measurement series at the wall, radiation exposure of $(0.296 \pm 0.017)\mu\,\mathrm{Sv}$ is obtained. In addition, a threshold voltage of $(512 \pm 2)\,\mathrm{V}$ was measured for the Geiger counter, and a voltage of $\geq 524\,\mathrm{V}$ was reached in the plateau region. The zero effect of the Geiger counter for the measurement in the lab is $(0.750 \pm 0.057)\,\mathrm{Imp\,s^{-1}}$ and $(0.333 \pm 0.032)\,\mathrm{Imp\,s^{-1}}$ for the measurement inside the lead shielding. The Geiger counter statistics of the long-term measurement with a uranium oxide tile show that the decays are Poisson distributed and approach a normal distribution for long gate times and thus large expected values. The responsiveness of the Geiger counter is $(4.45 \pm 0.41)\,\%$.

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1 Introduction

Radioactivity is a property of atomic nuclei. The decay of these nuclei is a stochastic process and has a broad spectrum of applications. For instance, the determination of the age of organic matter via the half-life or the application in medicine. In the latter, controlled radioactive radiation is used to destroy cancer cells. Not only this shows the omnipresent importance of radioactivity. Since the Chernobyl reactor accident in 1986, large quantities of radioactive cesium-137 have reached Germany. The effect can be seen in the form of increased radiation exposure in animals and fungi. Radioactive radiation also comes with risks. Namely, it has a harmful effect on biological tissue, which can be quantified by the equivalent dose. This dose can be measured using a dosimeter, which is done for cesium-137. The influence of the distance to the radioactive substance on the equivalent dose is checked in the following experiment by holding the dosimeter at a distance of 1 m above the radioactive substance. Furthermore, a Geiger-Müller counter is used to determine the half-life of barium. This Geiger counter is also used to examine the impulse probability distribution of a uranium-oxide-covered brick and whether it follows a normal distribution or Poisson distribution. Finally, the zero effect is investigated, to find out the influence of radioactivity on the environment.

2 Theory

The radioactive decay is completely random, but a statistical statement about the probability of the decay can be made if the number of atoms is high. With the following differential equation

$$A = \lambda N = -dNdt, \tag{1}$$

the radioactivity of the nucleus is calculated. Here λ is the decay constant, which describes the proportionality between the size of a group of radioactive atoms and the rate at which the group decreases due to radioactive decay. This differential equation gives the number of decays per second and can be used to derive the decay law

$$N(t) = N_0 e^{-\lambda t} = N_0 e^{-\frac{\lambda}{\tau}}. \tag{2}$$

Here $N(t)$ is the number of atomic nuclei not yet decayed at time t and N_0 is the number of radioactive nuclei at time $t = 0$. Furthermore, $\tau = 1/\lambda$ is the mean life time. From equation (2), the half-life can be determined

$$T_{1/2} = \frac{ln(2)}{\lambda}. \tag{3}$$

Radioactive decay can be classified into three types of decay. In α-decay ${}^{4}_{2}He^{2+}$ particles are emitted from the nucleus. These are also called α-particles and have a large ionization power which is dangerous for humans because it can cause dangerous cell mutations during DNA replication. However, the α-particles have a comparatively low penetration depth. In β-decay, a neutron of the nucleus converts into a proton. In this process, either an electron β^- or an positron (proton converts to neutron) β^+ is emitted. The ionization power of this radiation is smaller than for the α-particle but the penetration power is larger. The γ-decay is not a decay but rather a byproduct of the α-decay and β-decay. The reason for this is that after these two decays the nucleus is in an excited state which wants to be dissolved. This happens by the emission of gamma radiation. Since this is electromagnetic radiation, gamma radiation has by far the largest penetration depth of the three radiation classes. Gamma radiation has three different interaction processes which cause the electromagnetic waves to vanish at some point. The first one is the photoelectric effect. This process dominates the other two processes for energies up to 1 M eV. Within an energy spectrum of 1 M eV to 5 M eV Compton scattering is the dominating process. For energies above 5 M eV pair production is the dominating process. These interactions increase with the nuclear mass of the absorbing material.
Cesium 137 for instance decays by β-decay into the meta-stable isotope barium 137m. By emitting a gamma-ray barium 137m eventually relaxes into the stable isotope barium 137.

3 Evaluation

In the following section, the radiation properties of cesium-137 and barium 137m are examined. Furthermore, the Geiger counter and the statistical distribution of the decay of a tile coated with uranium oxide is investigated.

3.1 Measurement with the dosimeter

In this part, the radiation properties of cesium-137 are examined. The cesium is integrated into a metal pin. This pin blocks the beta radiation emitted by the cesium. In addition, the metal pin is screwed into a lead block, so that no gamma radiation escapes. If a plug attached to the top of the lead block is removed, a collimated gamma beam escapes. To measure the dose rate of the gamma beam, a series of measurements is made with the Neva dose rate meter at a distance of one meter and directly on the surface of the lead block. The error of the best value is the standard error resulting from the variance of the measurement series. Thus the dose rate is

$$\dot{D_{0m}} = (4.25 \pm 0.12)\,\mu\,Sv\,h^{-1}$$
$$\dot{D_{1m}} = (0.193 \pm 0.030)\,\mu\,Sv\,h^{-1}.$$

If we now extrapolate the absorbed doses to a period of four hours we get

$$D_{0m} = (18.80 \pm 0.49)\,\mu\,Sv$$
$$D_{1m} = (0.193 \pm 0.030)\,\mu\,Sv.$$

Thus, within four hours a fraction of $(0.3616 \pm 0.0099)\,\%$ of the annual limit of 5 m Sv is detected at the surface of the lead cube and $(0.155 \pm 0.0024)\,\%$ of the annual limit of 5 m Sv is detected at a distance of one meter. Here it becomes clear that a distance of one meter to the radiation source reduces the absorbed dose significantly. In this distance to the cube one could stay $(25.8 \pm 4.0) \cdot 10^3$ h long until the annual limit is reached.

3.2 Radiation exposure in the lab

To measure the radiation exposure during the experiment, the RADEX RD 1503 dosimeter, which is attached to the wall, measures the local dose rate every 15 minutes. Subsequently, the measured values are averaged. The series of measurements yields a total averaged radiation dose of $(0.296 \pm 0.017)\mu$ Sv. In one year (8760 h), the radiation exposure in the lab has an absorbed dose of (942 ± 52) m Sv.

3.3 Geiger-Müller counter characteristics

The gas discharge in an ionization chamber depends on the applied voltage, as the voltage amplifies the ion flow. The current flow increases according to Ohm's law until saturation occurs. In the saturation zone, no greater ionization can be achieved even by a high absorbed radiation dose, since the cross-section exceeds the mean free path of the ions. In the saturation zone, the current flow depends only on the incident absorbed radiation dose. The Geiger-Müller counter, which is physically based on an ionization chamber, also has a plateau region. In addition, it has a threshold for the voltage. Below the threshold, no pulses can be measured with the Geiger counter. To determine the threshold of the Geiger counter, it was held above the lead cube with the cesium. The voltage was increased until the first pulses were recorded with the Leybold digital meter. The threshold is (512 ± 2) V. The error of the applied voltage results from the fact that the Leybold digital meter can only increase the voltage in 4 V steps. To determine the plateau zone of the Geiger counter, the number of pulses at a gate time of 60 s is measured for voltages between the determined threshold of (512 ± 2) V and 600 V. These are then plotted in figure 1. To localize the plateau

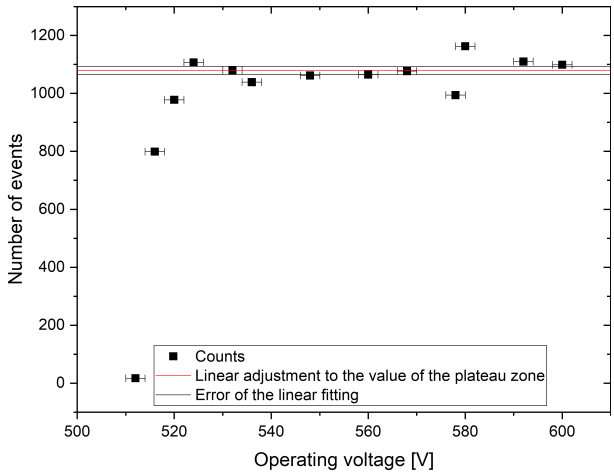

Fig. 1: The response of the Geiger Müller counter for different operating voltages. The threshold is (512 ± 2) V and the plateau zone begins at voltages of $\geq$ 524 V.

zone, a constant with its corresponding standard error was placed in the plateau zone. The first value which takes at least the value of this constant is at a voltage of (524 ± 2) V. Thus, the plateau zone begins for voltages of $\geq$ 524 V. For further measurements, the voltage is set to the value of (580 ± 2) V, which is inside the plateau zone too.

3.4 Zero effect

The zero effect is the response of the Geiger counter in the absence of the radiation source. This effect has two causes. One is the dark current which is caused by spontaneous formation of free charge carriers and the other is the background radiation of the lab. To quantify the zero effect, a series of measurements with a gate time of 60 s is performed for the Geiger counter in the lab and the Geiger counter in a lead shield. So, the values of further measurements with the Geiger counter can be adjusted for a zero effect of

$$ZeroEffect_{unshielded} = (0.750 \pm 0.057)\, Imp\, s^{-1}$$
$$ZeroEffect_{shielded} = (0.333 \pm 0.032)\, Imp\, s^{-1}.$$

The values for the number of pulses are only physically valid in the form of integers. Therefore, the zero effect can only be included in a measurement for gate times greater than 1 s and must be rounded to integers.

3.5 Half-life

After the zero effect has been determined, the radiation behavior of the unstable barium 137m can be observed with the Geiger counter. The barium isotope is dissolved from the cesium with a 0.4 normal hydrochloric acid. The activity of the solution is measured with the Geiger counter and then plotted graphically against the time (Figure 2).

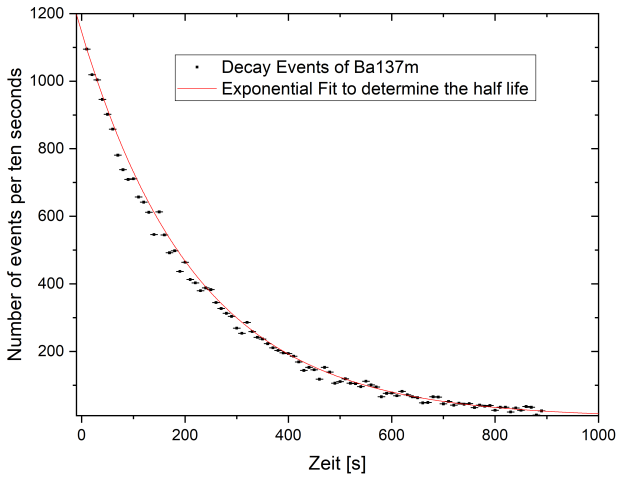

Fig. 2: The decreasing number of decay events. An exponential fit yields a half-life of $T_{1/2} = (154.3 \pm 3.1)$ s. The sender is eluted barium 137m.

An exponential function of the form

$$f(t) = Ae^{at} + c$$

can be applied to the data set. The parameter a is using (2)

$$a = \frac{ln(2)}{T_{1/2}}$$
$$T_{1/2} = \frac{ln(2)}{a}.$$

The exponential fit yields $a = (44.90 \pm 0.90)\, 10^{-4}\, s^{-1}$. This yields a half-life of

$$T_{1/2} = (154.3 \pm 3.1)\, s.$$

This value is consistent with the literature value of 153.12 s [1].

3.6 Counting statistics

In this part of the experiment, the statistical distribution of the decay of a tile coated with uranium oxide is observed. Therefore, the decay rate of a tile in a lead shield is measured with the Geiger counter. The measurements are performed for gate times of 1 s and 10 s at a voltage of 600 V. Furthermore, it is checked whether the decays can be described with the Poisson distribution. Thus, a χ^2 test is performed. For the gate time of 10 s it is also checked if a normal distribution can describe the data.

3.6.1 Gate time 1 s

For a gate time of 1 s, 1738 measurements were taken. The mean value of this measurement is $\mu = (0.652 \pm 0.019)$ Imp with a standard error determined from the measurement series. The following null hypothesis is used: "The impulses measured under the above conditions follow a Poisson distribution." To test this null hypothesis, the measured values are divided into classes (Appendix). The frequency of a class is determined according to the Poisson distribution and must be ≥ 5. Classes with a lower frequency are merged until they meet this condition. This results in five classes and thus three degrees of freedom for a Poisson distribution. To accept the null hypothesis with a 5 % probability of error we expect a value of $\chi^2 \leq 7.81$ [2]. From the table given in the Appendix follows $\chi^2 = 2.75$. Thus, the null hypothesis can be accepted and the decay rate at a gate time of 1 s follows a Poisson distribution. The measured frequency of the classes at a gate time of 1 s is shown together with the expected frequency according to the Poisson distribution in Figure 3.

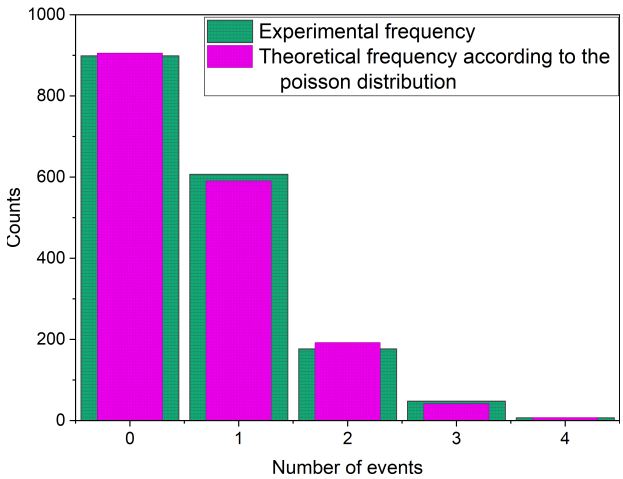

Fig. 3: Histogram for the number of decay events measured with the Geiger counter at a working voltage of 600 V and with a gate time of 1 s. The sender is a uranium oxide-coated tile.

3.6.2 Gate time of 10 s

For the gate time of 10 s, 466 measurements were taken. The mean of this measurement is $\mu = (6.89 \pm 0.12)$ Imp with the corresponding standard error. The following null hypothesis is used: "The impulses measured under the above conditions follow a Poisson distribution." As in the previous section, the measured values are divided into classes (Appendix). With an error probability of 5 %, the result for the 13 classes and 11 degrees of freedom is $\chi^2 \leq 19.68$. From the table in the appendix, the value $\chi^2 = 13.13$ is used. Thus, the measured values are Poisson distributed with an error probability of 5 %. The measured frequency of the classes at a gate time of 10 s is shown together with the expected frequency according to the Poisson distribution in Figure 4. A Poisson distributed quantity can generally

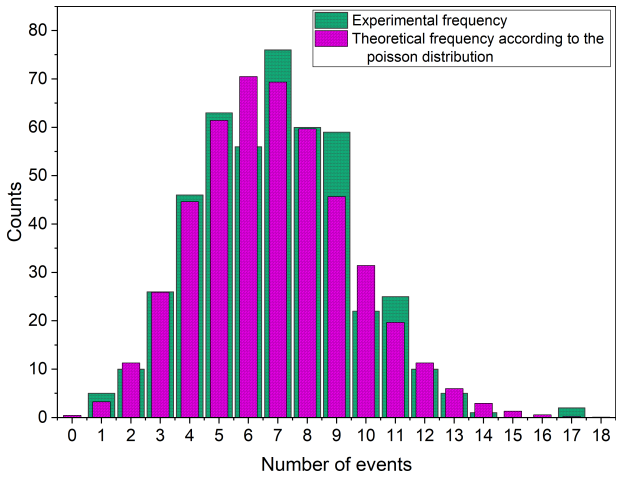

Fig. 4: Histogram for the number of decay events measured with the Geiger counter at a working voltage of 600 V and with a gate time of 10 s. The sender is a uranium oxide-coated tile.

follow a normal distribution for an expected value of $\mu \geq 9$. This is not satisfied for our measurement. Even though for this case the normal distribution is not optimal to describe the data, it can be used as an approximation. The quality of this approximation becomes clear when the cumulative probability of the individual classes is plotted on a probability axis in the y-direction (Figure 5). For a normally distributed

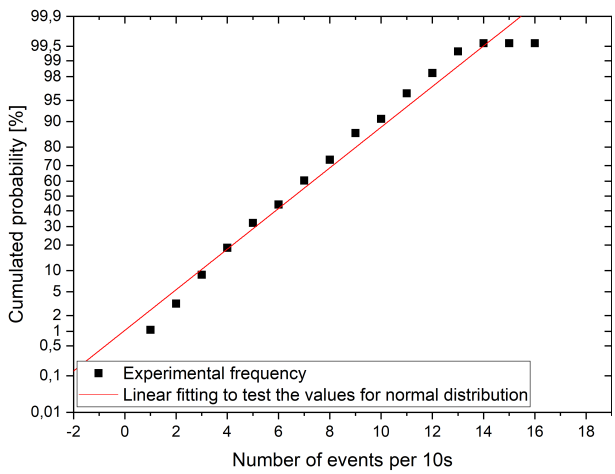

Fig. 5: The cumulative probability of the measurements with a gate time of 10 s. The linear fit shows that the normal distribution provides an approximation to the data.

random variable, one expects a linear fit with a correlation coefficient of $r = 1$. The correlation coefficient $r = 0.98809$ of the fit in figure 5 shows that the normal distribution describes the measurement well although significant deviations from the line are observed. These deviations are due to the expectation value being ≤ 9.

3.7 Responsiveness of the Geiger-Müller counter

To check which fraction of the emitted gamma quanta is detected by the Geiger counter, the expected activity of the cesium preparation must first be determined. For this purpose, it has to be considered when and with which activity the preparation was delivered. According to the information in the lab, the used preparation S772 was delivered on August 06, 1966, with an activity of 3.70 MBq. The activity A shows how many decays per second occur in the preparation. Since the number of decays is proportional to the number of particles, the decay law (2) together with the half-life (3) gives

$$A(t) = A_0 e^{\frac{ln(2)}{T_{1/2}} t}.$$

With a half-life for cesium of $\mathrm{T}_{1/2} = 30.2\,\mathrm{a}$, the expected activity is

$$A_{expected} = 1.04505\, M\, Bq.$$

The counting tube has an area of $S = 0.0011\,\mathrm{m}^2$ and was radiated at a distance of $\mathrm{d} = 0.085 \pm 0.005\,\mathrm{m}$ from the cesium. A series of measurements yields a activity of

$$A_{measured} = (47.94 \pm 3.4)\, Bq.$$

This value is adjusted for the zero effect. Taking into account the solid angle, the ratio of $\mathrm{A}_{\mathrm{expected}}$ and $\mathrm{A}_{\mathrm{measured}}$ gives

$$\frac{A_{measured}}{A_{expected}} \cdot \frac{4\pi d}{S} = (4.45 \pm 0.41)\,\%.$$

The determined value for the detected fraction of gamma rays shows that a large part of the gamma quanta was not detected with the counting tube. This is because we assume a point source. This approximation has a significant influence for small distances. Furthermore, it can happen that single gamma quanta hitting a gas molecule does not ionize it. Therefore, there is no current surge, which can be measured by the Geiger counter.

4 Conclusion

Cesium 137 is shielded by an aluminum bar and a lead cube. Directly at the lead cube, at a distance of one meter and at the wall of the lab the local absorbed dose rate of the preparation is measured. The short-time measurement series for the dose rates yields $\dot{\mathrm{D}}_0 = (4.25 \pm 0.12)\,\mu\,\mathrm{Sv\,h^{-1}}$ and $\dot{\mathrm{D}}_{1\mathrm{m}} = (0.193 \pm 0.030)\,\mu\,\mathrm{Sv\,h^{-1}}$. The long-term measurement series on the wall yields a radiation exposure of $(0.296 \pm 0.017)\mu\,\mathrm{Sv}$. Within four hours a fraction of $(0.3616 \pm 0.0099)\,\%$ of the annual limit of $5\,\mathrm{m\,Sv}$ is recorded at the surface of the lead cube and $(0.155 \pm 0.0024)\,\%$ of the annual limit of $5\,\mathrm{m\,Sv}$ at a distance of one meter. Here it becomes clear that a distance of one meter to the radiation source already reduces the absorbed dose significantly what complies with the initial theoretical considerations. In this distance to the cube, one could stay $(25.8 \pm 4.0)10^3\,\mathrm{h}$ long until the annual limit is reached. In addition, a threshold voltage of $(512 \pm 2)\,\mathrm{V}$ was measured for the Geiger counter, and a voltage of $\geq 524\,\mathrm{V}$ was reached in the plateau region. The zero effect of the counting tube is $\mathrm{ZeroEffect}_{\mathrm{unshielded}} = (0.750 \pm 0.057)\,\mathrm{Imp\,s^{-1}}$ for the measurement in space and $\mathrm{ZeroEffect}_{\mathrm{shielded}} = (0.333 \pm 0.032)\,\mathrm{Imp\,s^{-1}}$ for the measurement inside the lead shielding. The Geiger counter statistics of the long-term measurement with a uranium oxide tile show that the decays are Poisson distributed and approach a normal distribution for long gate times and thus large expected values. Finally, it was tested which fraction of the emitted gamma quanta is detected by the Geiger counter. A value of $(4.45 \pm 0.41)\,\%$ was determined for the responsiveness of the Geiger counter.

References

[1] Das Periodensystem der Elemente online, http://www.periodensystem-online.de/index.php?id=isotopeel=56mz=137nrg=0.6617show=nuklidsel=, last visit 17.09.2021

[2] Data and Error Analysis Lecture notes, Second Chapter

5 Appendix

k	f_{Exp}	f_{Theo}	χ^2
0	899	905.59	0.047955587
1	607	590.36	0.469018226
2	177	192.43	1.237254586
3	48	41.81	0.916433867
≥ 4	7	7.81	0.084007682
$\sum$	1738	1738	2.754669949

Table 1: The measurement of the uranium oxide coated tile with a gate time of 1s, sorted in classes depending on the number of events. Moreover, the table shows the experimental frequencies as well as the poisson frequencies together with the $\text{Chi}^2 - deviations$.

k	f_{Exp}	f_{Theo}	χ^2
≤ 2	15	15.05	0.000166113
3	26	25.91	0.000312621
4	46	44.61	0.043310917
5	63	61.44	0.039609375
6	56	70.52	2.989653999
7	76	69.37	0.633658642
8	60	59.71	0.001408474
9	59	45.69	3.877349529
10	22	31.46	2.844615385
11	25	19.7	1.425888325
12	10	11.3	0.149557522
13	5	5.99	0.163622705
≥ 14	3	5.25	0.964285714
$\sum$	466	466	13.13343932

Table 2: The measurement of the uranium oxide coated tile with a gate time of 10s, sorted in classes depending on the number of events. Moreover, the table shows the experimental frequencies as well as the poisson frequencies together with the $\text{Chi}^2 - deviations$.